



1.9

Rational Functions and Vertical Asymptotes

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: A noncanceled denominator zero is a vertical-asymptote candidate.

2. Write one precise sentence explaining this principle: Odd multiplicity changes the denominator sign across the asymptote; even multiplicity does not.

3. Write one precise sentence explaining this principle: The numerator sign near the asymptote also matters.

4. Write one precise sentence explaining this principle: One-sided statements are stronger and more precise than saying only "the function goes to infinity."

Core

5. Describe one-sided behavior of $\frac{1}{x-2}$ at $x = 2$.
6. Describe one-sided behavior of $-\frac{3}{(x+1)^2}$ at $x = -1$.
7. Why must common factors be canceled before classifying denominator zeros?
8. State what the formula $\lim_{x \rightarrow a^-} R(x)$, $\lim_{x \rightarrow a^+} R(x)$ tells you and name any required restriction.
9. State what the formula $Q(a) = 0$, $P(a) \neq 0 \Rightarrow x = a$ is a vertical asymptote tells you and name any required restriction.
10. Create a second representation (table, graph, formula, or context) for one central relationship in Rational Functions and Vertical Asymptotes. State its domain and justify one key feature.

Medium

11. Find every vertical asymptote and state both one-sided behaviors for $\frac{2}{x-5}$.
12. Find every vertical asymptote and state both one-sided behaviors for $\frac{-3}{(x+2)^2}$.
13. Find every vertical asymptote and state both one-sided behaviors for $\frac{x+1}{(x-3)^3}$.

14. Find every vertical asymptote and state both one-sided behaviors for $\frac{x-4}{(x+1)^2}$.

15. Find every vertical asymptote and state both one-sided behaviors for $\frac{(x-2)^2}{x+4}$.

Hard

16. Analyze $\frac{x-1}{(x+2)(x-3)}$ near each vertical asymptote.

17. Find k so $\frac{x+k}{(x-2)^2}$ tends to $+\infty$ on both sides of $x=2$.

18. Can a rational function cross a vertical asymptote? Explain.

19. After simplifying $\frac{(x-1)^2}{(x-1)^3(x+2)}$, classify $x=1$ and $x=-2$.

Difficult

20. Construct a rational function with vertical asymptote $x=1$ where both sides go to $+\infty$, and $x=-2$ where left goes $+\infty$ and right goes $-\infty$.

21. For $R(x) = \frac{x^2+k}{(x-1)^2}$, determine k values for which $x=1$ is a vertical asymptote with both sides negative.

22. Explain how the parity of a denominator factor and the sign of the remaining factors determine one-sided divergence.

Challenge / HOT

23. Can a reduced rational function have infinitely many vertical asymptotes? Prove.

24. Construct a reduced rational function with vertical asymptotes at every real zero of $x^4 - 5x^2 + 4$, and prescribe alternating one-sided signs.

AP-Style Multiple Choice

25. Before finding vertical asymptotes, first
(A) differentiate
(B) simplify and identify remaining denominator zeros
(C) find horizontal asymptote
(D) set numerator equal to zero
26. At an even-power denominator zero with positive numerator, both sides tend to
(A) $+\infty$
(B) $-\infty$
(C) opposite infinities
(D) 0
27. $1/(x+3)$ as $x \rightarrow -3^-$ tends to
(A) $+\infty$
(B) $-\infty$
(C) 0
(D) 1
28. A canceled denominator factor creates
(A) always a vertical asymptote
(B) a hole if fully canceled
(C) a zero
(D) a horizontal asymptote
29. Which is excluded from domain?
(A) horizontal asymptote y-value
(B) vertical asymptote x-value
(C) slant asymptote points
(D) all zeros
30. $-2/(x-1)^2$ has
(A) opposite infinities
(B) both + infinity
(C) both - infinity
(D) a hole

AP-Style Free Response

31. Analyze $R(x) = \frac{(x+1)(x-4)}{(x-2)^2(x+3)}$. For each vertical asymptote, state both one-sided limits and justify with a sign chart.

32. Let $R_k(x) = \frac{x+k}{(x-1)^3}$. Classify the one-sided behavior at $x=1$ for $k>-1$, $k=-1$, and $k<-1$.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.